

Joseph Barnett

Tech Architect ·
Distinguished Engineer

CONTACT

+1.617.447.7697
joseph.barnett@gmail.com
Walla Walla, Washington
linkedin.com/in/josephbarnett

SKILLS

AI & AI Infrastructure

AI cost attribution & observability,
LLM gateways & telemetry proxies,
agentic systems & multi-agent
orchestration, AI-augmented
engineering

Kubernetes & Cloud-Native

Deep Kubernetes (EKS,
CRI/CNI/CSI, RBAC, runtime
security, GitOps), Docker, IaC
(Terraform, Pulumi,
CloudFormation), multi-cloud (AWS,
Azure, GCP)

Observability & Operations

Telemetry pipelines, metrics &
tracing, cost attribution, CI/CD,
incident response & runbooks, SRE
practices

Systems & Internals

Linux/Windows OS internals,
firmware, kernel & driver
development, virtualization design,
system architecture, distributed
systems

Networking

Protocol design & stacks (MPLS,
RTSP, routing), global edge &
Anycast, datacenter-to-cloud scale-
out, eBPF telemetry, VPC/CIDR &
egress design

Security & Compliance

FedRAMP, SOC 2, ISO 27001,
application security, enterprise key
management & revocability

Data & Stream Processing

Snowflake, Trino, stream
processing, Postgres, MS SQL
Server, Oracle, DynamoDB,
MongoDB

Languages & Compilers

Compiler & DSL design and
implementation, language runtimes
& interpreters; Go, Rust, Python,
Java, C#, C++, JavaScript, ASM

SUMMARY

Distinguished engineer and architect with 30 years building pioneering technology — from kernel and network protocol stacks to cloud-native platforms and AI-era observability. Drive large architectural decisions end-to-end: design, ship, support, and validate in production. Operate as a force multiplier across engineering, product, sales, and the C-suite, turning deep technical capability into market-defining business outcomes.

PROFESSIONAL EXPERIENCE

CloudZero / Architect, Distinguished Engineer

MAY 2024 – PRESENT · WALLA WALLA, WA

- Architecting the **Collector and AI Gateway** — capturing both infrastructure and AI spend and tying it to the business value it produces, the market's hardest current problem. The foundation that redirected the company's pivot to its new **AI ROI / Financial Control Plane** category.
- Took the AI-telemetry agent from prototype to production in **two weeks**, catalyzing a company-wide P0.
- Architect and owner of CloudZero's **Kubernetes cost-attribution capabilities** — open-source policy and effort definitions, first shipped software, and strategy.
- Scaled cost attribution to **50 TB/week** across **10,000+ node** clusters and **5,000+ clusters** under observation — serving enterprises including **DoorDash** and **Twilio** while **halving the solution's cost** for both customers and CloudZero.
- Reduced Kubernetes cost-attribution error from **20%+ to within 0.09%**.
- Mentored engineers and emerging leaders across the organization — coaching up, down, and across was a core part of the role.

Slim.AI / First Engineer, Distinguished Engineer

NOV 2020 – MAY 2024 · ACTON, MA

- Spearheaded a container minification and security platform, contributing to a Series A raise and an early market buyout offer.
- Designed and built research prototypes including a Docker Desktop Extension that delivered 20% month-over-month user growth.
- Architected the foundational CI/CD, platform, workflow-orchestration, security, and core scanning/intelligence/build engines that landed the company's first customer.
- Engaged directly with design partners to align product with market needs; traveled internationally for on-site onboarding and design sessions.

Cisco Systems / Product Architect, Lead Security Architect, AppSec Manager

JAN 2017 – NOV 2020 · WALTHAM, MA

- Led cloud-security engineering and architecture for a transformation achieving FedRAMP, SOC2, and ISO 27001 compliance across 63 teams.
- Partnered with global sales to secure a \$5M investment into the cloud-security division to drive compliance programs.
- As Federal Customer Engineer, drove adoption through design and procurement engagements that helped open a \$1B federal market opportunity for Cisco Umbrella.
- Delivered conference presentations establishing thought leadership in cloud-native security and compliance.

Metratech / Ericsson / Software Engineer VI DEC 2009 – JAN 2012 · WALTHAM, MA

- Led a domain-specific-language billing-platform generation engine for high-speed inline billing and contract enforcement.
- Led design and delivery of the billing system for the Microsoft Azure platform.
- Drove system refactoring and integration through the Ericsson acquisition, ensuring a seamless transition.

Egenera, Inc. / Sr. Software Engineer JAN 2007 – DEC 2009 · MARLBOROUGH, MA

- Built virtualization and management technologies for server, chassis, rack, and warehouse control, improving data-center efficiency.

Motorola / Software Engineer III FEB 2003 – JAN 2007 · SOUTHBOROUGH, MA

- Engineered OS support packages, drivers, and firmware for Linux kernel distributions on the ATCA product line.

Netplane Systems / Software Engineer I–II JUN 2000 – FEB 2003 · DEDHAM, MA

- Built the first commercial video-on-demand platform — for NTT DoCoMo, over an MPLS backbone using a JavaScript interpreting engine and RTSP.
- Developed cross-platform backbone protocol stacks that helped capture 70% of the telecommunications protocol market.
- Delivered advanced engineering training on networking and routing protocols.

EDUCATION

Northeastern University — B.S., Computer Engineering — *magna cum laude* · 2002–2007 · Boston, MA

Portland Community College — A.S., Computer Software Engineering Tech. — *magna cum laude* · 1997–2000 · Portland, OR

COMMUNITY

Active open-source contributor and collaborator across industry peer efforts; regular participant in community hackathon events.